

Learning-Rate Warmup and Schedules

How the step size should change over a run · video 2:16

This lesson answers one question: should the learning rate stay fixed during training? The same tiny GPT is trained for 3,000 steps with five schedules. The constant learning rates end near 1.60; schedules that **decay** the learning rate end lower (cosine 1.551, linear 1.554), and **warmup-stable-decay** (WSD) does best (1.542), with most of its gain arriving in the short decay at the end.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 5).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. Five schedules, and warmup	▶ 0:08-0:52	2	☐
2. Constant vs decaying	▶ 0:52-1:23	2	☐
3. Warmup-stable-decay	▶ 1:23-1:59	3	☐

Before you start: Deep Dive episodes 16 (warmup and stability) and 19 (AdamW). Code: `code/d20_lr_schedules` (PyTorch only; five tiny models, about 25 minutes on a CPU).

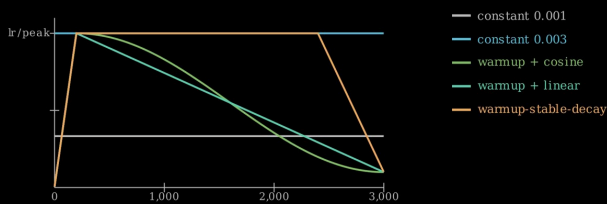
CONCEPT 1 OF 3

Five schedules, and warmup

▶ Video 0:08-0:52

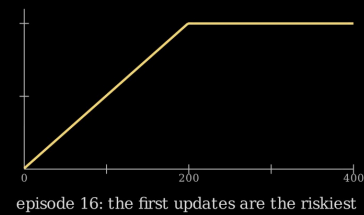
Almost no large model keeps its learning rate fixed: it rises at the start and falls toward the end. The experiment trains the same tiny GPT for 3,000 steps, five times: two **constant** learning rates (0.001 and 0.003) and three schedules peaking at 0.003, all ending at 10% of the peak: warmup + **cosine** decay, warmup + **linear** decay, and **warmup-stable-decay** (WSD): warmup, a long stable phase at the peak, then a short linear decay over the last 600 steps.

the same tiny GPT, 3,000 steps, five learning-rate schedules



▶ 0:39 Five schedules: constant 0.001 and 0.003; cosine, linear and warmup-stable-decay, all peaking at 0.003.

warmup: 200 steps from near zero to the peak



▶ 0:51 Warmup: 200 steps from near zero to the peak.

Each schedule starts with **warmup**: 200 steps rising linearly from near zero to the peak. As episode 16 showed, the first updates are the riskiest, and keeping them small protects the model.

KEY IDEA

A schedule is a multiplier of the peak learning rate as a function of the step: warm up, hold, decay.

Check 1: Five schedules, and warmup

Answer all 2 questions before looking at the answers. Pass mark: all correct.

1.1 CALCULATION

With 200 warmup steps and peak 0.003, what is the learning rate at step 99 (the code uses $(\text{step} + 1) / 200$)?

.....

1.2 CALCULATION

At step 1,600 (halfway through the decay), what are the cosine, linear and WSD learning rates?

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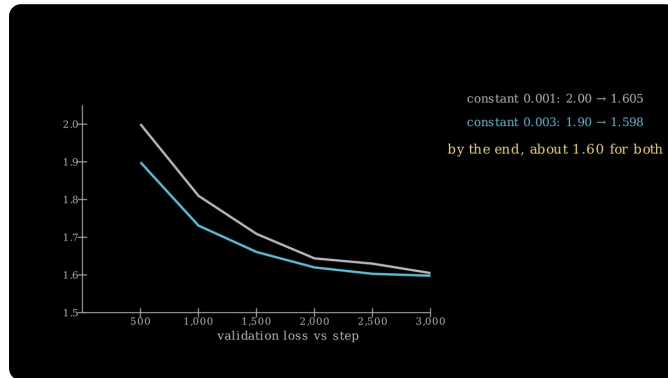
CONCEPT 2 OF 3

Constant vs decaying

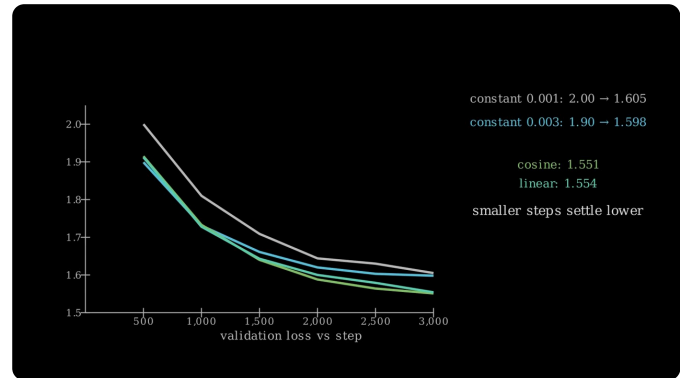
▶ Video 0:52-1:23

The higher constant learning rate learns faster early: validation loss 1.90 at step 500 against 2.00. But by the end they nearly meet: **1.598** and **1.605**. A fixed step size eventually stops helping.

The decaying schedules pull ahead as the learning rate shrinks: cosine **1.551**, linear **1.554**. One way to picture it: smaller steps let the model settle into a lower point that big steps keep jumping over.



▶ 1:07 Constant learning rates: 0.003 is faster early (1.90 vs 2.00 at step 500); both end near 1.60.



▶ 1:22 Cosine and linear decay pull ahead as the learning rate shrinks: 1.551 and 1.554.

KEY IDEA

A high learning rate makes fast early progress; a decaying one lets the loss settle lower at the end.

Check 2: Constant vs decaying

Answer all 2 questions before looking at the answers. Pass mark: all correct.

2.1 TRUE OR FALSE

"The constant learning rate of 0.003 ended clearly better than 0.001."

True ☐ False ☐

2.2 CALCULATION

What is the cosine schedule's learning rate at step 2,700?

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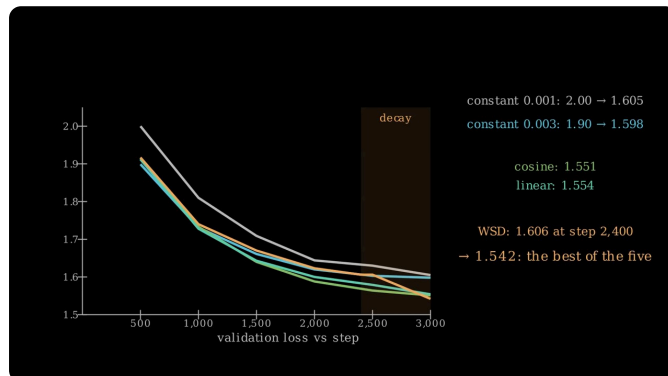
CONCEPT 3 OF 3

Warmup-stable-decay

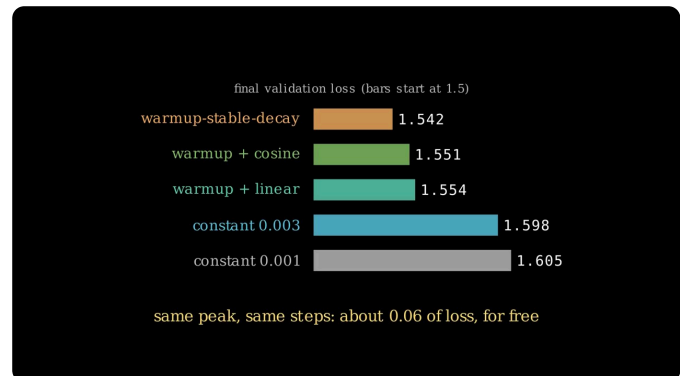
▶ Video 1:23-1:59

For most of the run, WSD stays at the peak, and its loss tracks the constant run: **1.606** at step 2,400. Then, in the last 600 steps, the learning rate falls to 10% of the peak and the loss drops sharply to **1.542**, the best of the five.

That shape is practical: you can keep training at the peak for as long as you like, and decay only when you want a finished model. The schedule matters, not just the peak value: about **0.06** of loss here (1.598 → 1.542), with the same peak and the same number of steps.



▶ 1:41 WSD tracks the constant run (1.606 at step 2,400), then drops to 1.542 in the 600-step decay.



▶ 1:57 Final losses: WSD 1.542, cosine 1.551, linear 1.554, constant 1.598 and 1.605.

KEY IDEA

Much of the benefit of decay arrives during the decay itself: WSD gains 0.064 in its last 600 steps.

Check 3: Warmup-stable-decay

Answer all 3 questions before looking at the answers. Pass mark: all correct.

3.1 PUT IN ORDER

Order the five schedules from best to worst final loss: *constant 0.003 · cosine · WSD · constant 0.001 · linear*.

3.2 SHORT ANSWER

Why is WSD convenient if you don't know in advance how long you will train?

3.3 HANDS-ON CODE

Try it yourself. Evaluate the episode's WSD function at a few steps.

```
for s in (0, 99, 1600, 2700, 2999):  
    print(s, 3e-3 * SCHEDULES["warmup-stable-decay"](s))
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. Five schedules, and warmup · rewatch ▶ 0:08-0:52

1.1 0.0015. 100 / 200 of the peak.

1.2 0.00165, 0.00165 and 0.003. Cosine and linear are both at 55% of the peak halfway; WSD is still in its stable phase until step 2,400.

2. Constant vs decaying · rewatch ▶ 0:52-1:23

2.1 False. 1.598 vs 1.605: it was ahead early (1.90 vs 2.00 at step 500), but they nearly met.

2.2 About 0.000376. $\text{warm} \times (0.1 + 0.45 (1 + \cos(\pi \cdot 2,500 / 2,800))) \times 0.003$, computed with the episode's function.

3. Warmup-stable-decay · rewatch ▶ 1:23-1:59

3.1 WSD (1.542) → cosine (1.551) → linear (1.554) → constant 0.003 (1.598) → constant 0.001 (1.605). From the episode's run.

3.2 Its learning rate does not depend on the total length until the final decay: you can keep training at the peak and start the decay whenever you decide to stop. Cosine and linear schedules need the total number of steps from the start.

3.3 0.000015, 0.0015, 0.003, 0.00165, 0.000304. Warmup to step 199, stable until 2,399, then a linear decay to 10%.